

Effects of Resistance Training Overload Progression Protocols on Strength and Muscle Mass

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ABSTRACT

The aim of this study was to compare the effects of progressive overload in resistance training on muscle strength and cross-sectional area (CSA) by specifically comparing the impact of increasing load (LOADprog) versus an increase in repetitions (REPSprog). We used a within-subject experimental design in which 39 previously untrained young persons (20 men and 19 women) had their legs randomized to LOADprog and REPSprog. Outcomes were assessed before and after 10 weeks of training. Muscle strength was assessed using the one repetition maximum (1RM) test on the leg extension exercise, and the CSA of the vastus lateralis was assessed by ultrasonography. Both protocols increased 1RM values from pre (LOADprog: 52.90 ± 16.32 kg; REPSprog: 51.67 ± 15.84 kg) to post (LOADprog: 69.05 ± 18.55 kg, REPSprog: 66.82 ± 17.95 kg), with no difference between them ($P > 0.05$). Similarly, both protocols also increased in CSA values from pre (LOADprog: 21.34 ± 4.71 cm²; REPSprog: 21.08 ± 4.62 cm²) to post (LOADprog: 23.53 ± 5.41 cm², REPSprog: 23.39 ± 5.19 cm²), with no difference between them ($P > 0.05$). In conclusion, our findings indicate that the progression of overload through load or repetitions can be used to promote gains in strength and muscle hypertrophy in young men and women in the early stages of training.

Introduction

Resistance training (RT) is considered the most effective method to increase muscle strength and cross-sectional area (CSA) (i. e.

muscle hypertrophy) [1]. For these adaptations to occur continuously, the RT program must be systematically modified, compelling the human body to adapt to the changing stimuli [2, 3]. It is

recommended that these alterations in RT stimulus align with three general principles of training progression: 1) progressive overload, 2) variation, and 3) specificity [2, 3]. Among these principles, progressive overload aims to gradually increase stress on the body during RT [1], which can be promoted by manipulating one or more RT variables such as intensity (load [kg]), volume (number of sets and/or repetitions), frequency (number of RT sessions per week), duration of repetitions (duration of the eccentric, concentric and isometrics phases) and pause between sets (rest interval) [1]. Traditionally, the progressive overload is performed by increasing the load to a repetition maximum zone (LOADprog), in which the load is adjusted when the individual performs more repetitions than the upper limit of the repetition zone (e. g. 9 to 12 repetitions) [1]. However, the LOADprog has been challenged by studies that have investigated the progression of overload by increasing the number of repetitions (REPSprog) performed until concentric muscle failure [4–6]. In this model, individuals can perform as many repetitions as possible until muscle failure. To date, little is known whether the REPSprog promotes different gains in muscle strength and mass compared to the LOADprog.

Recently, the effects of LOADprog and REPSprog in previously untrained individuals on muscle strength and CSA were compared through a secondary and exploratory analysis of previously published data [7]. The results indicated greater strength gains and muscle hypertrophy for LOADprog compared to REPSprog. Furthermore, significant correlations were observed between the progression of volume load (VL = sets x repetitions x load [kg]) and strength gains as well as muscle hypertrophy. However, the lack of randomization is a limitation for attributing causality for the greater effects observed in LOADprog compared to REPSprog. Therefore, the hypothesis that LOADprog can produce greater VL progression, gains in strength and muscle hypertrophy requires further scrutiny.

The primary aim of the present study was to compare the effects of LOADprog and REPSprog protocols on muscle strength and muscle CSA. We hypothesized that LOADprog would promote greater gains in strength and muscle hypertrophy compared to REPSprog. As a secondary aim, we compared the effects of LOADprog and REPSprog on VL and the association between VL progression with strength gains and muscle hypertrophy. We hypothesized that LOADprog would promote greater VL progression than REPSprog and that there would be an association between VL progression and gains in muscle strength and muscle hypertrophy.

Materials and Methods

Participants

Thirty-nine participants (20 men and 19 women) between 18 to 35 years old who had not practiced RT, aerobic training or any other physical training program for at least 6 months (age: 24 ± 4 years, body mass: 68.2 ± 12.4 kg, height: 1.70 ± 0.08 meters, BMI: 23.5 ± 3.7 kg/m², relative strength on the knee extension exercise: 1.53 ± 0.34 kg/bm) volunteered to participate. Individuals who had musculoskeletal injuries or any neuromuscular disorder of the lower limbs were excluded. To calculate the sample size, we performed a power analysis based on simulations for two-way repeated-measures ANOVA. Using the Superpower package in RStudio [8], we ran

10,000 simulations and established that a minimum sample size of 38 participants was necessary to achieve a statistical power of 80 %, considering $\alpha = 0.05$, $r = 0.72$ (correlation between legs) and an effect size for an ordinal interaction of $f = 0.12$ (Cohen's f statistic). The correlation value between legs was obtained from previous studies in our laboratory for the increase in one repetition maximum test (1RM) values ($r = 0.66$) and muscle CSA ($r = 0.78$) [6, 9–13]. Considering possible dropouts, we chose to start the study with a larger number of participants. All subjects were educated about the potential risks and benefits, and provided written informed consent after full disclosure of methods. This study was approved by the ethics committee of the University (5.505.441) and was conducted in accordance with the Declaration of Helsinki [14].

Experimental design

We used a prospective, randomized controlled, single-blind (researchers blinded to the assessments) and intrasubject study (i. e. each leg performed a training protocol) conducted between July and October 2022. This study was reported following the recommendations described in the Consolidated Standards of Reporting Trials (CONSORT) checklist and used the CONSORT extension for within-subject designs [15]. The option for the within-subject design was due to the balanced dropout rates to be equal between the experimental conditions and makes, and removes the between-subject variability from the statistical analysis observed in parallel group designs (e. g. energy intake, training history, sleep and individual biologic factors) [15, 16]. In this sense, each participant's leg was randomly allocated to the: i) load progression model (LOADprog) or ii) repetitions progression model (REPSprog). The sequence of allocation of each leg to the RT protocols was performed using a stratified block randomization (i. e. strata leg dominance and strength asymmetry between legs) [17]. To determine the strength asymmetry, we calculated the percentage difference in 1 RM values between legs. This procedure was performed for all participants, resulting in an ordered list of leg differences. Next, the list was divided into three parts. Individuals belonging to the first tertile had the highest values of differences representing the high strata, while the second and third tertiles represented the moderate and low strata, respectively. The allocation sequence was obtained through an internet randomization service [18]. After generating the allocation code, the RT protocol to which each leg of the participants was allocated was communicated directly by the researcher to the other researchers responsible for applying the RT. In this study, the LOADprog protocol was used as a positive control group because it is traditionally performed and recommended to promote gains in strength and muscle hypertrophy [1, 19].

In the first laboratory visit, vastus lateralis CSA from both legs was assessed using ultrasonography. After 72 hours, vastus lateralis CSA were assessed again to determine the typical error (TE) and coefficient variation (CV) of the assessment. On that same day, the leg extension exercise 1RM was performed on both legs. Similarly, to CSA, the 1RM test was performed after 72 hours. The RT was performed for 10 weeks. Finally, the ultrasound and the 1RM test were performed 96 hours after the last RT session. The study protocol was retrospectively registered in Registry of Clinical Trials.

Muscle cross-sectional area

Muscle CSA was obtained from the vastus lateralis through B-mode ultrasound (US) images using a 7.5 MHz linear probe (MySono U6, Samsung), following the procedures validated by Lixandrão et al. [20]. Participants were instructed to refrain from vigorous physical activity for at least 72 h before each assessment [21, 22]. Before the assessments, participants laid down for 15 minutes to ensure fluid redistribution. Then, the evaluator measured the distance between the greater trochanter and the lateral epicondyle of the femur to determine the length of the right and left femur. The point corresponding to 50 % of the length was marked with a dermatographic pen as a reference point for image acquisition. From the 50 % point, the skin was marked in the medial and lateral directions every 2 cm to guide the displacement of the probe in the sagittal plane, parallel to the long axis of the femur. Water-soluble transmission gel was applied to ensure acoustic coupling of the US probe without compressing the epidermis. To obtain the muscle CSA, the US probe was displaced in the sagittal plane, starting at the alignment point of the upper edge of the probe with the most medial skin mark (over the rectus femoris muscle) and ending on the lateral side of the thigh. Then, the images were opened in the software Power Point (Microsoft, USA) and manually rotated to reconstruct the entire fascia of the vastus lateralis muscle to allow panoramic visualization of the entire muscle CSA [23]. The reconstructed image was opened in ImageJ software, and the “polygonal” function was used to circle and measure the CSA of the vastus lateralis, excluding the underlying muscle fascia and bone tissue as much as possible. The typical error (ET) and the coefficient of variation (CV) were 0.60 cm² and 2.83 %, respectively.

Muscle strength

Muscle strength was measured using the 1RM performed unilaterally on a leg extension machine (Effort NKR; Nakagym) following the recommendations described by Brown et al. [24]. Initially, the participants performed a general warm-up on an ergometric bicycle at 20 km · h⁻¹ for 5 minutes, followed by two sets of specific warm-ups on the leg extension chair. In the first set, the participants performed eight repetitions with approximately 50 % of the 1RM. Next, three repetitions were performed with a load of approximately 70 % of the 1RM. A two-minute break was applied between warm-up sets. After warming up, participants performed the 1RM test. In the leg extension machine, the participants started the test with the knee flexed (90° verified by a manual goniometer), then performed the concentric phase of the movement up to the maximum comfortable amplitude, and finally returned to the initial position. Participants had up to five attempts to reach their 1-RM load, with a rest of 3 min between attempts. The ET and CV were 1.67 kg and 3.27 %, respectively.

Resistance training protocols

The RT protocols encompassed the unilateral leg extension exercise performed two to three times per week for 10 weeks. Although the weekly frequency of RT may have varied between individuals over the weeks, all individuals completed 23 training sessions. In addition, the experimental design adopted allowed these variations not to influence the comparison between protocols, since both legs of each subject were trained on the same day, with a 2-min rest interval between them. Finally, to mitigate potential

cross-protocol influences, the sequence of protocol execution was randomized and counterbalanced between sessions. At the beginning of each RT session, participants performed a general warm-up on a cycle ergometer (Ergo-Fit, Pirmasens, Rheinland-Pfalz, cycling at 20 km · h⁻¹ for 5 min).

LOADprog was performed for four sets of 9–12 repetitions maximum and 90 seconds of rest between sets. The load was adjusted set by set when concentric muscle failure occurred in a repetition outside the repetition maximum zone. The minimum value for exercise load increment was 1 kg. In the REPSprog protocol, the participants also performed four sets with 80 % of 1RM (i. e. fixed load throughout the training program) until concentric muscle failure. Load increment occurred whenever the participant was able to perform more repetitions until muscle failure. Concentric muscle failure was considered the inability to perform another concentric repetition while maintaining proper form. Verbal encouragement was provided for both protocols to ensure maximum effort. The absolute and relative load (i. e. percentage of 1RM) of the first and last session were used to compare the load of the protocols at the beginning and end of the experimental period. The sum of the VL of all sessions was used to calculate the accumulated VL of each RT protocol. The VL of each session was used to compare the progression of the VL.

Blinding

To avoid the risk of bias related to the measurement of outcomes, researchers who performed the 1RM test and ultrasound were blinded to training protocol assigned to each leg. Study participants were instructed and frequently reminded not to disclose leg allocation to evaluators. Due to the characteristics of the study, it was not possible to blind the participants and trainers. Additionally, participants were blinded to the study hypothesis. They were also instructed not to engage in any other type of physical training and/or behavioral changes that could affect members differently. In addition, trainers were previously instructed on the protocols applied to ensure the accuracy of the prescription and avoid any differences between protocols.

Statistical analyses

Data normality for all dependent variables was verified using Shapiro-Wilk test, and variance homogeneity was verified using Levene tests. A mixed model analysis with protocols and time as fixed factors and participants as random factors was used to compare primary (1RM and CSA) and some secondary outcomes (absolute and relative load used in the first and last session). In the case of a significant *F* value, Tukey's adjustment was used for multiple comparisons. Secondary outcomes, such as load, number of repetitions, area under the curve (AUC) of VL progression and the accumulated VL during the experimental period, were compared using paired *t*-test. The AUC analysis for VL progression was performed using the trapezium rule (GraphPad Prism, GraphPad Software, San Diego, CA, United States) in order to characterize the magnitude of VL changes over time. AUC analyses were calculated using the VL of all training sessions. Pearson's correlation were performed to investigate the association between VL progression and changes in 1RM and CSA for each protocol separately. The effect size and its respective 95 % confidence interval (CI) was interpreted and reported.

ed as the mean difference between groups in absolute values. The interpretation of this estimate allows a better practical interpretation of the results [25]. Data are presented as the mean $\pm$ standard deviation (SD). The significance level adopted was $P < 0.05$. Analyses were performed using RStudio and SAS 9.2 software (Institute Inc., Cary, NC).

Results

Participants

The initial characteristics of the participants stratified by protocols are described in ► **Table 1**. One hundred and forty-one participants were eligible for the study, but 53 participants agreed to participate and had their legs randomized to the protocols. Ten participants dropped out before starting the RT protocols. Forty-three started RT, but three discontinued consent for personal reasons. In addition, one participant withdrew consent prior to participating in the final assessments. Thus, 39 participants (i. e. 78 legs) were included in the analysis of primary and secondary outcomes. A CONSORT flowchart of the randomized controlled trial is shown in ► **Fig. 1**.

Muscle strength

The mixed model analysis demonstrated that there was no significant protocol vs. time interaction ($F = 1.71$; $P = 0.20$) or protocol main effect ($F = 3.21$; $P = 0.08$). However, a main effect of time was observed for the 1RM ($F = 213.6$; $P < 0.0001$) (► **Table 2**).

► **Table 1** Initial characteristics of each protocol.

Variables		LOADprog	REPSprog
1RM (kg)		52.9 $\pm$ 16.3	51.7 $\pm$ 15.8
CSA (cm ²)		21.3 $\pm$ 4.7	21.1 $\pm$ 4.6
Number of dominant legs		19 (48%)	20 (52%)
Number of strongest legs		24 (61%)	15 (39%)
Number of participants with leg imbalance in muscle strength	High	8	8
	Medium	6	6
	Low	5	6
LOADprog: overload progression protocol by load; REPSprog: overload progression protocol by repetitions; 1RM: one repetition maximum; CSA: muscle cross-sectional area. Data are presented as mean $\pm$ standard deviation.			

► **Table 2** Primary outcomes in the pre- and post-training assessments for resistance training with load progression (LOADprog) and with repetitions progression (REPSprog).

Outcomes	LOADprog			REPSprog			MD (CI 95%)
	Pre	Post	Δ Post-Pre	Pre	Post	Δ Post-Pre	
1RM (kg)	52.9 $\pm$ 16.3	69.0 $\pm$ 18.5*	16.1 $\pm$ 7.2	51.7 $\pm$ 15.8	66.8 $\pm$ 17.9*	15.1 $\pm$ 7.0	- 1.0 (- 2.6 to 0.6)
CSA (cm ²)	21.3 $\pm$ 4.7	23.5 $\pm$ 5.4*	2.2 $\pm$ 2.2	21.1 $\pm$ 4.6	23.4 $\pm$ 5.2*	2.3 $\pm$ 1.7	0.1 (- 0.5 a 0.8)
MD: Mean difference between groups. Negative values favor LOADprog; 1RM: 1 repetition maximum test; CSA: Muscle cross-sectional area; 95% CI: 95% confidence interval. Values reported as mean $\pm$ standard deviation. *Statistically significant difference compared to Pre (main effect of time).							

Muscle cross-sectional area (CSA)

Mixed model analysis revealed that there was no significant protocol vs. time interaction ($F = 0.02$; $P = 0.87$) or protocol main effect ($F = 0.57$; $P = 0.45$). However, a main effect of time was observed for CSA ($F = 94.07$; $P < 0.0001$) (► **Table 2**).

Repetitions

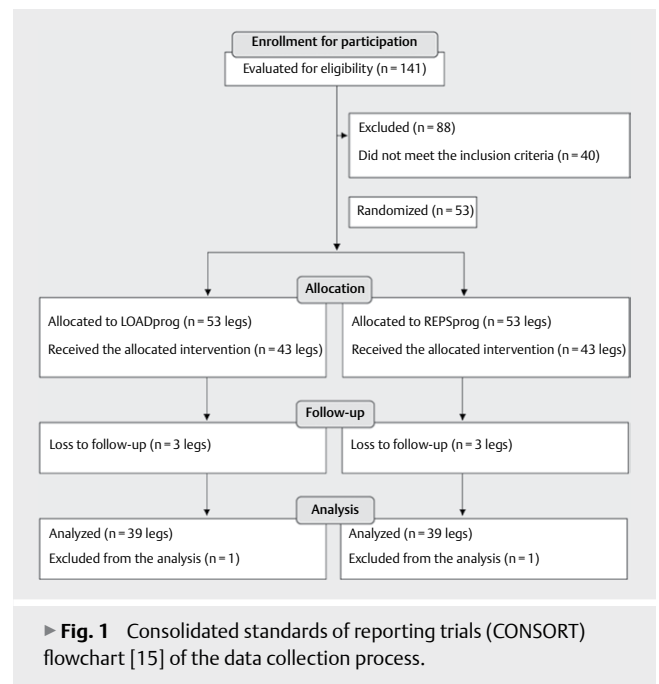
A paired t-test revealed that the number of repetitions accumulated during the RT program was significantly higher for REPSprog than for LOADprog (LOADprog: 1000 $\pm$ 67 reps; REPSprog: 1292 $\pm$ 302 reps; mean difference [95% CI]: - 292 reps [- 375 to - 207] $T = - 7.041$; $P < 0.0001$).

Load

A paired t-test revealed that the accumulated load of the RT program was significantly higher for LOADprog than for REPSprog (LOADprog: 1282 $\pm$ 392 kg; REPSprog: 989 $\pm$ 307 kg; mean difference [95% CI]: 293 kg [212 to 372]; $T = 7.384$; $P < 0.0001$).

Relative and absolute load of the first and last RT session

Mixed model analysis demonstrated a protocol vs. time interaction in the absolute and relative load ($F = 27.66$; $P < 0.0001$ and $F = 101.87$;



$P < 0.0001$, respectively) used in the first and last training sessions. The post hoc test indicated that the LOADprog protocol showed a significant increase in absolute (Pre: 42.3 ± 13.1 kg; Post: 61.4 ± 20.7 kg, Δ : 19.0 ± 12.9 kg, $P < 0.0001$) and relative (Pre: 80 % 1RM; Post: 88 % 1RM, Δ : $8.0 \pm 13.0\%$, $P < 0.0001$) load. For REPSprog, there was a significant decrease in the relative load (Pre: 80 % 1RM; Post: 62 % 1RM, Δ : $-18.0 \pm 7.0\%$, $P < 0.0001$), with no change in the absolute load (Pre: 41.6 ± 12.7 kg; Post: 41.6 ± 12.7 kg, Δ : 0 ± 0 , $P = 0.99$). Mean difference (95 % CI) between the protocols were 19 kg (14.8 to 23.2) and 26 % (20.7 to 31.7) favoring LOADprog.

Volume load (VL)

The paired *t* test revealed that the accumulated VL in the training sessions was not different between LOADprog and REPSprog (LOADprog: 53703 ± 17390 kg; REPSprog: 52528 ± 18283 kg; mean difference [95 % CI]: 1174 kg [-1289 to 3637]; $T = 0.96$; $P = 0.346$). Additionally, the paired *t* test did not indicate significant differences between the RT protocols in the AUC of VL progression ($T = 1.32$; $P = 0.19$) (► Fig. 2a, b).

Correlation between VL progression and strength gains and muscle hypertrophy

There was no significant correlation between VL progression and muscle strength gains for LOADprog ($R = 0.20$; CI 95 % [-0.11 to 0.50]; $P = 0.20$) or REPSprog ($R = 0.16$; CI 95 % [-0.16 to 0.46]; $P = 0.31$). Similarly, there was no significant correlation between VL progression and increases in CSA for LOADprog ($R = -0.04$; CI 95 % [-0.35 to 0.28]; $P = 0.80$) and REPSprog ($R = 0.08$; CI 95 % [-0.24 to 0.38]; $P = 0.64$).

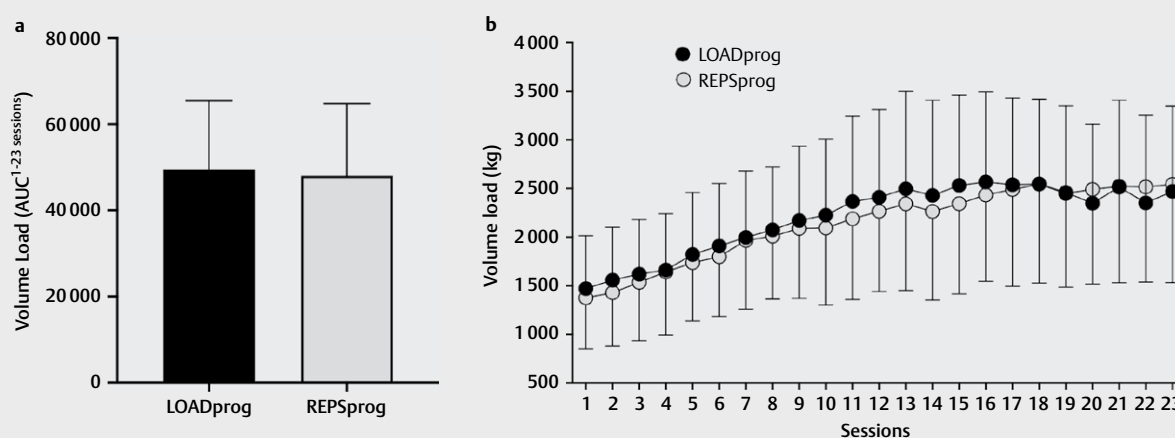
Discussion

The main findings of the present study were that there were no differences between LOADprog and REPSprog protocols on strength gains and muscle hypertrophy when implemented over a 10-week training period. These findings indicate that both overload progression protocols can be used by untrained young men and women to

increase strength and muscle hypertrophy in the early stages of training.

We observed an increase in 1RM values (LOADprog: 16.5 kg [32.8 %] and REPSprog: 15.5 kg [31.4 %]) and muscle CSA (LOADprog: 2.19 cm^2 [11.2 %] and REPSprog: 2.32 cm^2 [10.5 %]). The magnitude of the increase in strength and muscle hypertrophy is in line with previous studies performed with untrained men and women [26–30]. Although no significant differences were observed between protocols, the magnitude of effect was a mean difference of 1.0 kg (1.5 %) in the 1RM test favoring LOADprog. The confidence interval (CI) was 2.60 kg (5.1 %) favoring LOADprog and 0.60 kg (2.1 %) favoring REPSprog. For muscle CSA, the magnitude of the effect was a mean difference of 0.13 cm^2 (0.77 %), favoring REPSprog. The confidence interval (CI) was 0.76 cm^2 (2.03 %), favoring LOADprog and 1.03 cm^2 (3.57 %), favoring REPSprog. The magnitude of the effect as well as the CI suggest little or no effect comparing protocols. However, we understand that future studies investigating the minimal important difference for clinical practice or physical performance in muscle strength gains and hypertrophy could confirm this speculation. Importantly, the hypothesis of the present study was that the LOADprog protocol would produce greater gains in strength and muscular hypertrophy, which was based on a recent non-randomized study carried out from a secondary analysis [7]. However, the results of the present study did not confirm this hypothesis.

In the present study, although the initial relative load was the same for both RT protocols (i. e. 80 % 1RM), due to load progression, the LOADprog protocol ended the experimental period with a training load corresponding to 88 % of the 1RM test performed after training. On the other hand, in the REPSprog protocol, in which there was no load progression, the participants finished the RT program with a load corresponding to 62 %. However, it is possible that this reduction was not enough to provide differences in muscle strength gains between protocols. In fact, previous studies suggest that there seems to be a greater difference in strength gains between protocols with loads greater than 80 % 1RM when compared to protocols with loads below 60 % 1RM [31, 32]. It is



► **Fig. 2** (a) Area under the curve (AUC) of volume load progression for LOADprog and REPSprog protocols. (b) Session-by-session volume load progression for LOADprog and REPSprog protocols.

plausible to speculate that longer periods of REPSprog use would promote a greater decrease in relative load, affecting muscle strength gains. However, this hypothesis still needs to be tested. Regarding muscle hypertrophy, although the loads used in the protocols differed throughout the training, both models of overload progression were performed until concentric muscle failure. Thus, considering that a wide range of load and number of repetitions are equally effective in promoting muscle hypertrophy when the exercise is performed to muscle failure [31–35], it is not surprising that both protocols produced similar increases in muscle CSA. Indeed, our current data agree with those by Plotkin et al. [36] who compared the effects of LOADprog and REPSprog in trained subjects and did not observe differences in muscle strength gains and hypertrophy. Therefore, we suggest that both protocols to increase overload progression can be used to promote gains in muscle strength and hypertrophy.

In our study, we did not observe significant differences in accumulated VL between protocols. However, unlike Nóbrega et al. [7], we did not find differences in VL progression between LOADprog and REPSprog. It is possible that differences in the experimental design between the studies explain these findings. In the present study, we used a within-subject design, which allowed both protocols to be performed by the same subject, reducing the between-subject variability observed in parallel group designs. Thus, it is possible that the observed differences in VL progression between protocols are not a direct result of the protocols themselves. For this reason, we suggest that the overload progression protocols based on increasing the load or number of repetitions do not promote differences in the progression of the VL.

Another hypothesis tested in the present study is whether VL progression is associated with muscle strength gains and hypertrophy; however, this has not been confirmed. In fact, while some studies show that VL progression is aligned with muscle strength gains and hypertrophy [7, 10, 37], others suggest a lack of association between these variables [38, 39]. Although these results are difficult to reconcile, the differences between these findings can be explained in part by the difference between RT protocols, experimental design, and how VL progression is performed. We recognize that the association between two variables can be more complex than just a linear relationship. Therefore, the lack of correlations observed in the present study does not imply the absence of any form of association between the variables [40]. However, it is possible that the subject's ability to progress VL more or less is not associated with strength gains and muscle hypertrophy.

This study is not without its limitations, including the following: 1) We only investigated the hypertrophic responses of the vastus lateralis muscle. Thus, we cannot confirm that the results will be similar when investigating different muscle groups. Additionally, only a single point was assessed. Considering that non-uniform hypertrophy can occur within a single muscle, assessing multiple points would allow us to investigate how the different portions of the vastus lateralis muscle respond to different overload progression protocols; 2) Muscle strength assessments and RT were limited to the lower limbs, more specifically to the leg extension exercise, and should not be extrapolated to different exercises and muscle groups; 3) Our sample was exclusively comprised of previously untrained young individuals. We cannot ascertain whether differ-

ent adaptive behaviors may be observed in other populations (e. g. older adults, resistance-trained individuals, etc.); 4) Finally, the present study only involved 10 weeks of training. Therefore, we cannot say that our conclusions can be maintained over a longer period of intervention.

In conclusion, we did not find differences between the overload progression protocols by load and repetitions in strength gains and muscle hypertrophy in untrained individuals. These findings suggest that both progression protocols can be used for this population in the initial stages of training. Future studies should replicate and expand these findings by analyzing other muscle groups and longer training times.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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